THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE UNIT TEST SET C

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. In science, a model is: 1

(a) a guess that has not been tested
(b) an exact copy of the real object in every detail
(c) a simplified way of looking at a real system that keeps only what matters for the question
(d) a mathematical formula only

2. The magnifying glass on the page numbers of the textbook stands for: 1

(a) estimation **(b)** units of measurement
(c) careful observation **(d)** choosing direction

3. ‘Science does not need to be perfect to be reliable’. Which feature supports this? 1

(a) Only experts may question
(b) Facts never change
(c) Predictions that fail lead to better ideas
(d) Theories can never be wrong

4. The scattering of a beam of light by particles in a colloid is called the: 1

(a) photoelectric effect **(b)** Tyndall effect
(c) Doppler effect **(d)** Raman effect

5. 20 g of salt is dissolved in 80 g of water. The mass by mass percentage of the solution is: 1

(a) 25 % **(b)** 20 % **(c)** 30 % **(d)** 10 %

6. The process of forming pure crystals from a hot saturated solution on cooling is called: 1

(a) chromatography **(b)** sublimation **(c)** crystallization **(d)** distillation

7. The chlorine in NaCl is present as: 1

(a) Cl_2 molecule **(b)** Cl^+ ion
(c) Cl^- ion **(d)** neutral atom

8. The chemical formula of sodium sulfate is: 1

(a) Na_2SO_4 **(b)** $Na(SO_4)_2$ **(c)** $NaSO_4$ **(d)** Na_2S

9. On decomposing 18 g of pure water, the masses of hydrogen and oxygen obtained are: 1

(a) 4 g and 14 g **(b)** 9 g and 9 g **(c)** 1 g and 17 g **(d)** 2 g and 16 g

10. Cathode rays were shown by J. J. Thomson to be: 1

(a) streams of negatively charged particles **(b)** protons
(c) streams of positive ions **(d)** neutral waves

11. An atom of carbon has atomic number 6 and mass number 12. The number of neutrons in it is: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 12

12. The Greek philosophers who proposed indivisible particles called ‘atomos’ were: 1

(a) Leucippus and Democritus **(b)** Dalton and Proust
(c) Thomson and Rutherford **(d)** Bohr and Bury

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. A friend says, ‘A theory is just a guess.’ Respond in two sentences. 2

14. Why is the temperature always mentioned while stating the solubility of a substance? 2

15. Why did the balloon experiment (open flask) show a loss in mass while the closed set-up did not? 2

16. Why was the neutron needed to explain atomic masses? Give an example. 2

17. State two conclusions of Rutherford's gold foil experiment. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. Estimate the volume of air a person breathes in a day. Write the steps. 3

19. What is fractional distillation? Where is it used? 3

20. Students X and Y made oxides of copper in the ratios Cu : O = 4 : 1 and 8 : 2. Do these results justify the law of constant proportions? 3

21. Why are the symbols of elements written using Latin names? Give two examples with their norms. 3

22. Describe the process of distillation with the help of the apparatus parts and name one use. 3

OR

What do solubility curves show? State two inferences from a curve in which solubility of compound B increases sharply with temperature. 3

23. A compound has 40 % sulfur and 60 % oxygen by mass. How much oxygen is present in a sample with 20 g sulfur? 3

OR

Carbon monoxide has C : O = 3 : 4. How much oxygen combines with 9 g carbon? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** The solubility of two compounds A and B is plotted against temperature. A shows a small rise and B a large rise as the temperature increases from 20 °C to 80 °C. 5

(i) Which dissolves more at 60 °C?

(ii) If saturated solutions of equal mass of A and B at 80 °C are cooled to 40 °C, which deposits more crystals? Why?

(iii) Which compound can be purified by crystallization more efficiently?

OR

Case study: A farmer must prepare a spray from a liquid pesticide for rice. 5

(i) Why must the proportion be right?

(ii) If 1 mL is mixed to make 100 mL, find % v/v.

(iii) How much pesticide is needed for 20 L of spray at this strength?

(iv) Name the type of mixture.

25. (i) State the rules of Bohr–Bury for distribution of electrons. 5

(ii) Write the electronic configuration of Mg, Cl, S, K and Ca.

(iii) Give their valencies.

OR

(i) Describe Thomson's cathode ray tube experiment. 5

(ii) What did he conclude?

(iii) Describe the plum pudding model and why it was proposed.

END OF QUESTION PAPER — Total: 50 Marks